



Chapter 2

硅单晶生长及晶圆制造和清洗

Silicon Crystal Growth,

wafer manufacturing and cleaning

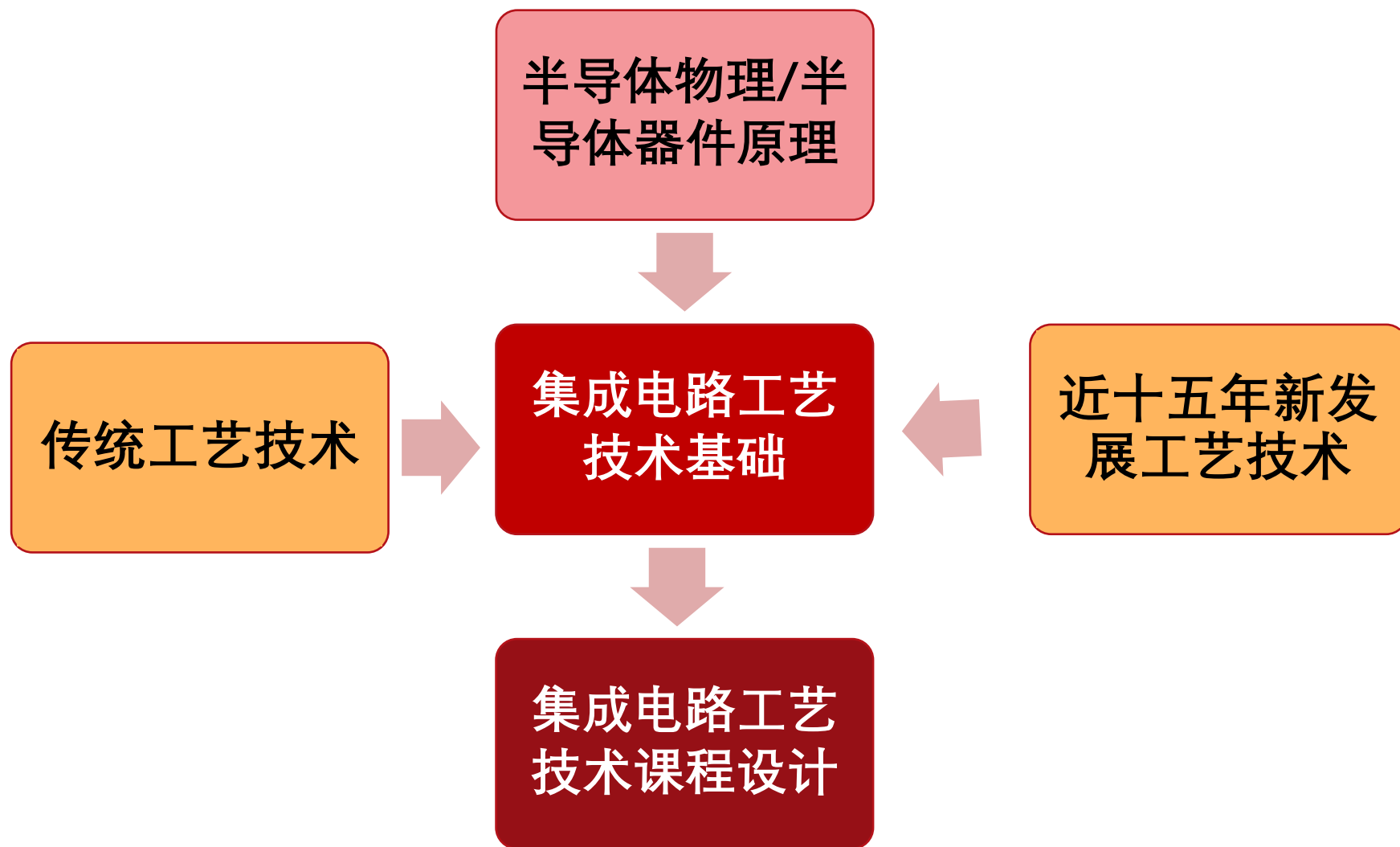


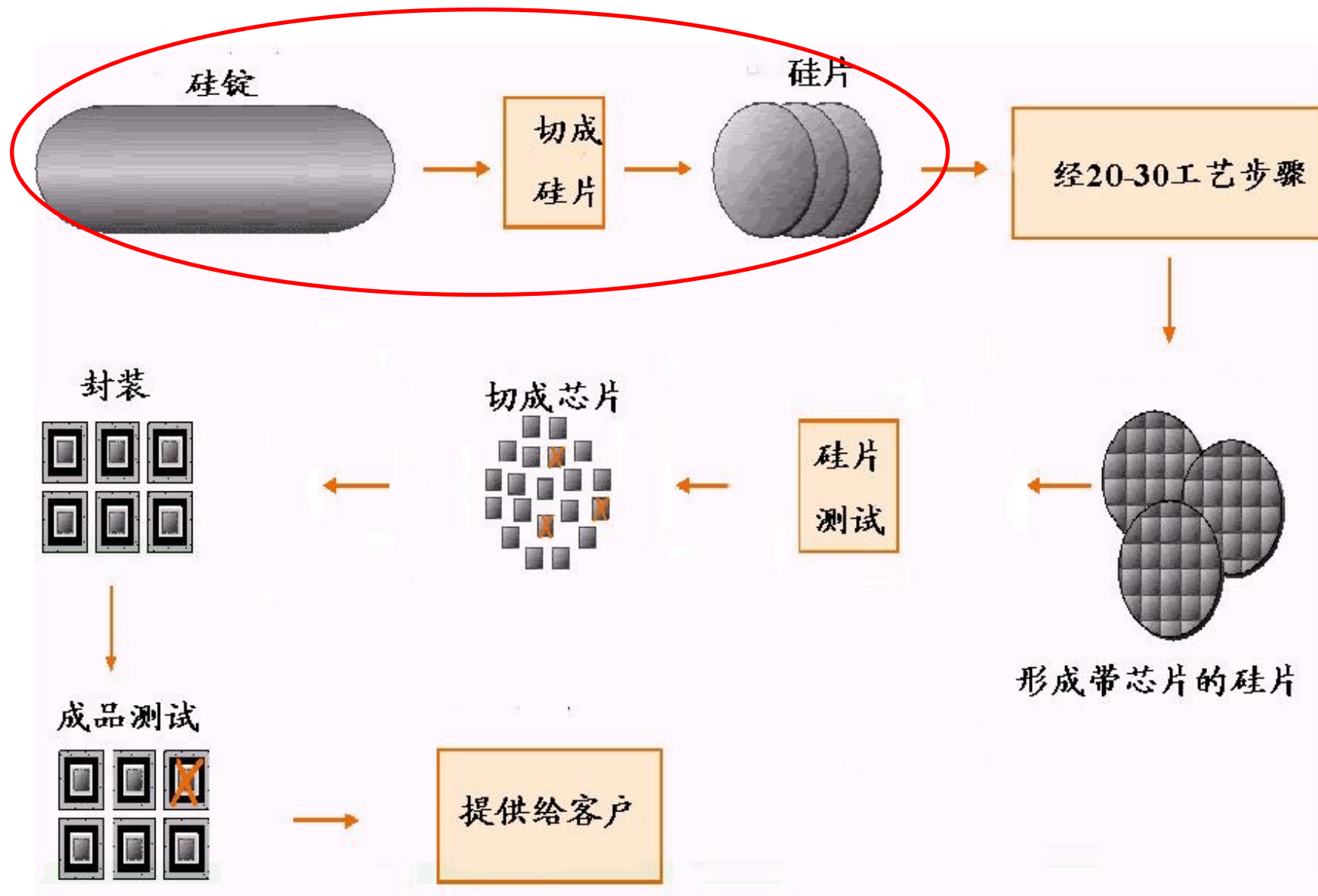
上海交通大学
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2025年11月13日





不同阶段经济价值对比



From sand to ultrapure Si single crystal wafers to CPUs



\$35 / ton

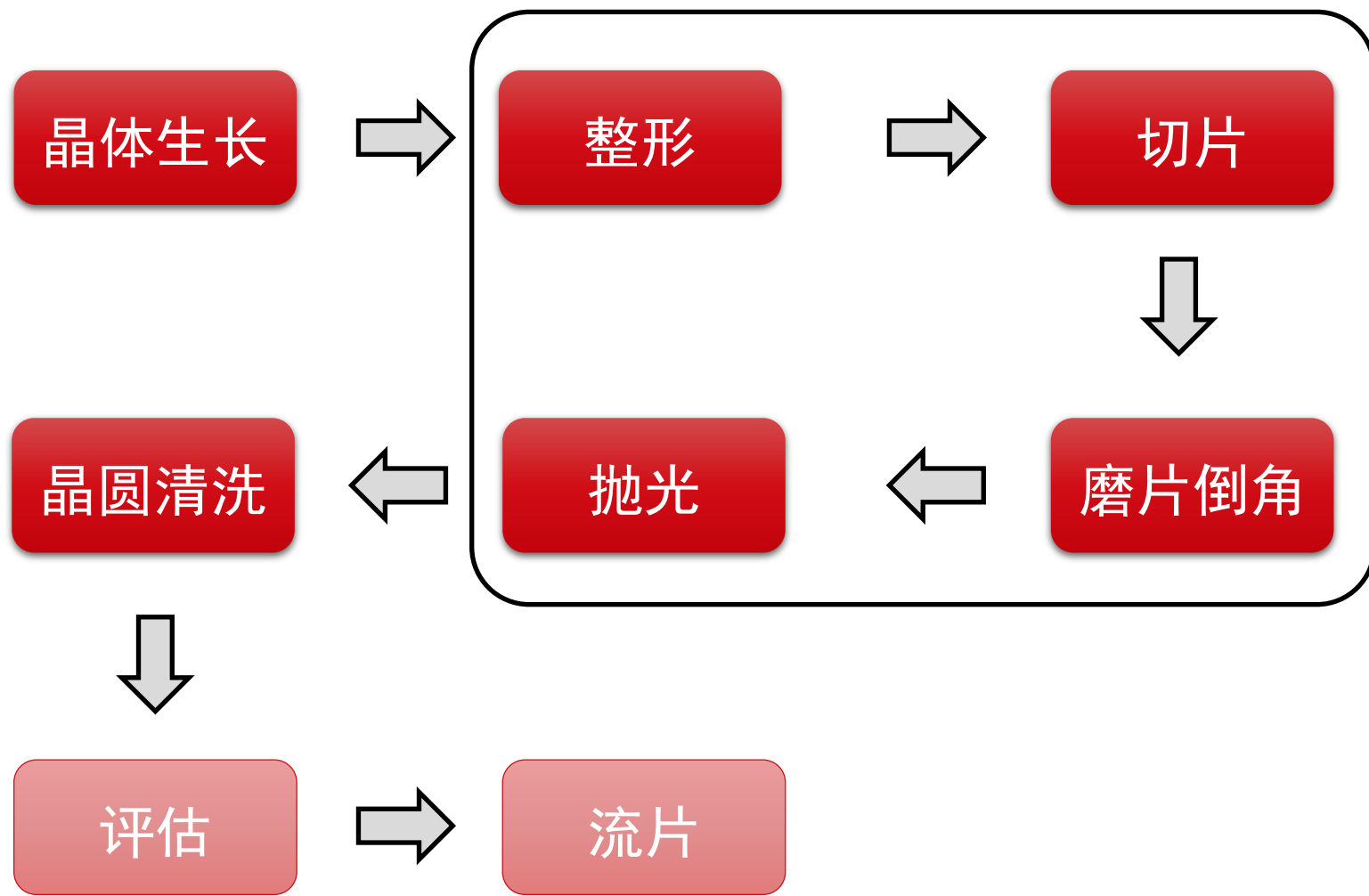
\$200 / 300mm wafer

\$300 / CPU

硅晶圆生长、制造流程



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✎ **2.1 Silicon crystal structure and defects.**

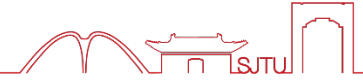
✎ **2.2 Single crystal growth.**

✎ **2.3 Wafer manufacturing**

✎ **2.4 Wafer cleaning**

✎ **2.5 Summary**

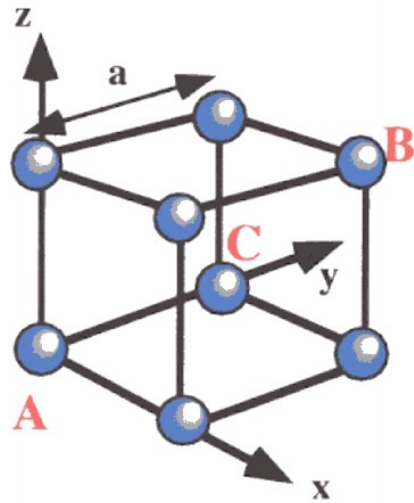
2.1.1 Review of crystal structure



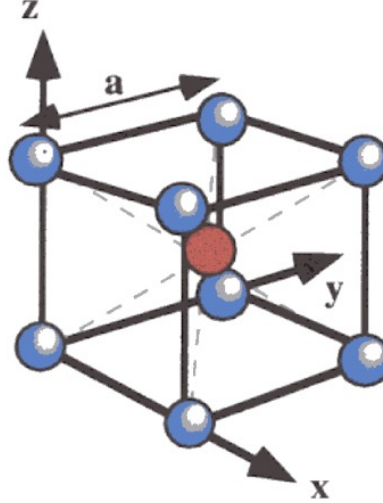
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Single-crystalline

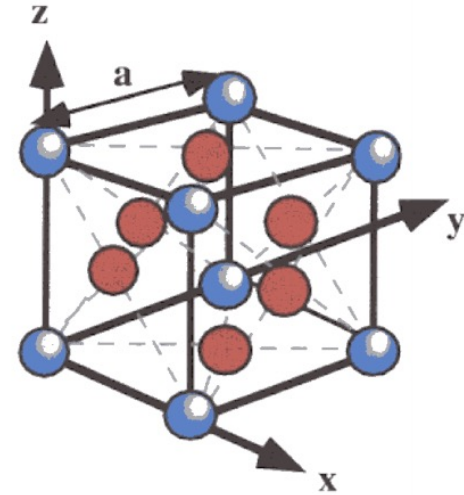
(A unit cell repeats throughout the solid, in the x, y, and z)



Cubic

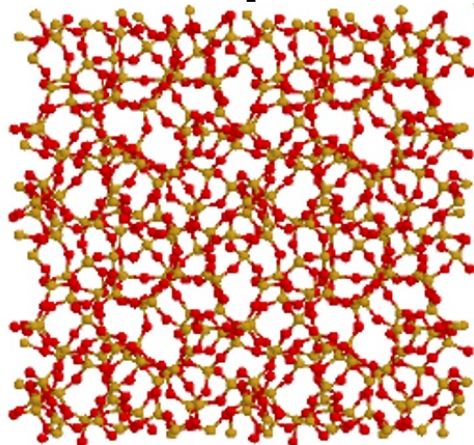


BCC

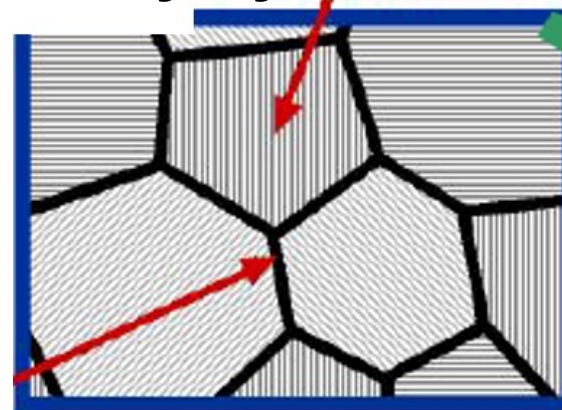


FCC

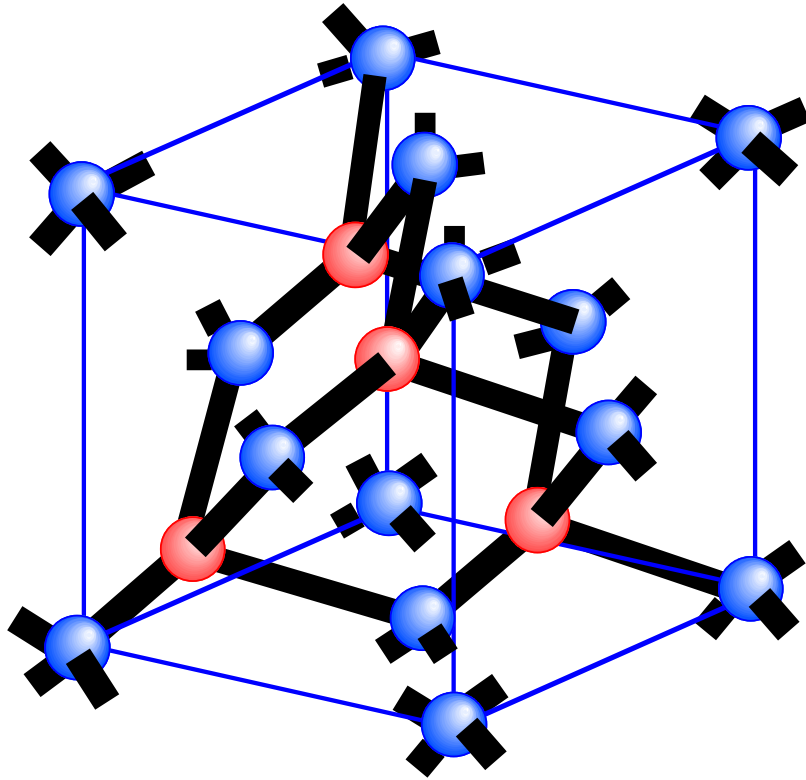
Amorphous



Polycrystalline

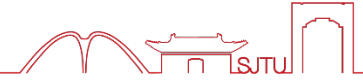


2.1.2 Dimond structure of Si

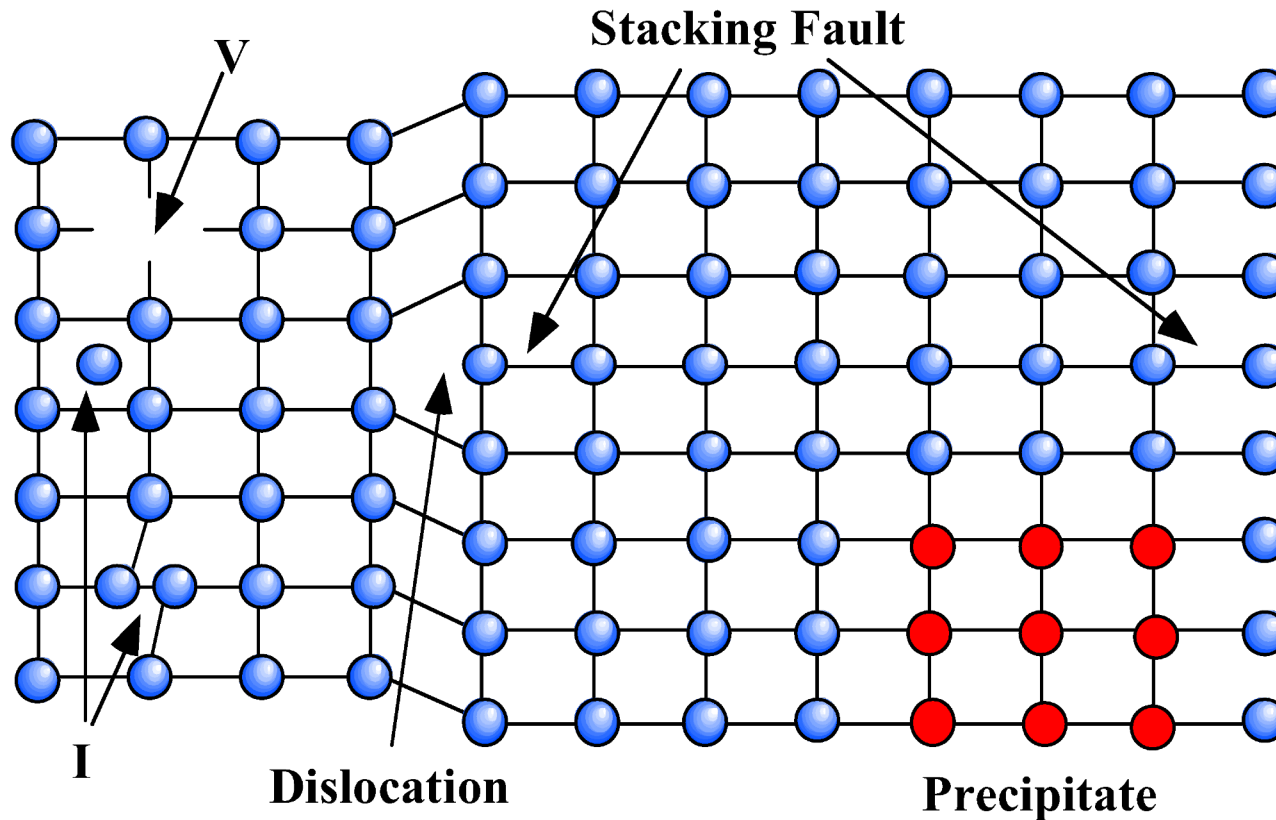


- Silicon has the basic diamond crystal structure
- Two merged FCC cells offset by $a/4$ in x , y and z .
- All atom are of the same type (Si)
- Atomic density:
 $N(\text{Si}) = 5 \times 10^{22} \text{ cm}^{-3}$

2.1.3 Defects in Si crystal



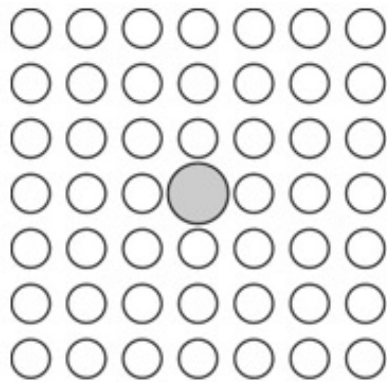
- Point defects: vacancy, interstitial, substitutional, Frankel defects
- Linear defects: edge and screw dislocations
- Planar defects: stacking fault, grain boundaries, surfaces
- Bulk defects: cracks and pores, precipitate



These defects are generally detrimental to device performance and also impact the behavior of processes (e.g. dopant diffusion)

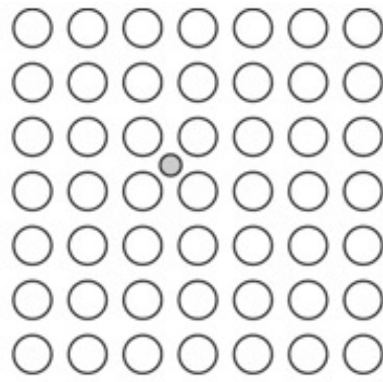
2.1.3 Point defects

Substitutional



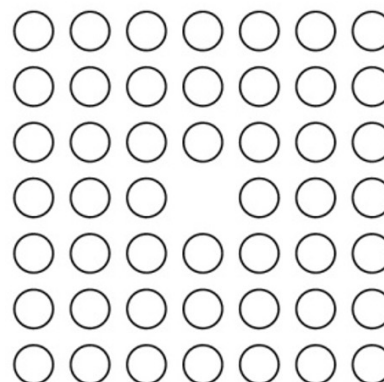
(a)

Interstitial



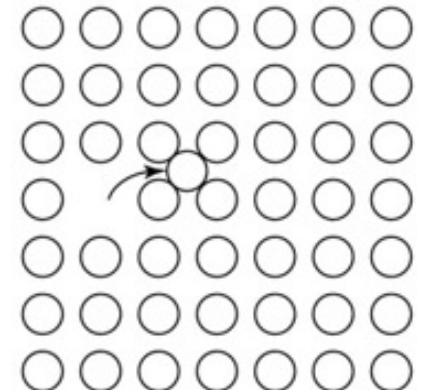
(b)

**Vacancy
Schottky defect**



(c)

**Frenkel-type
interstitial-vacancy pair**



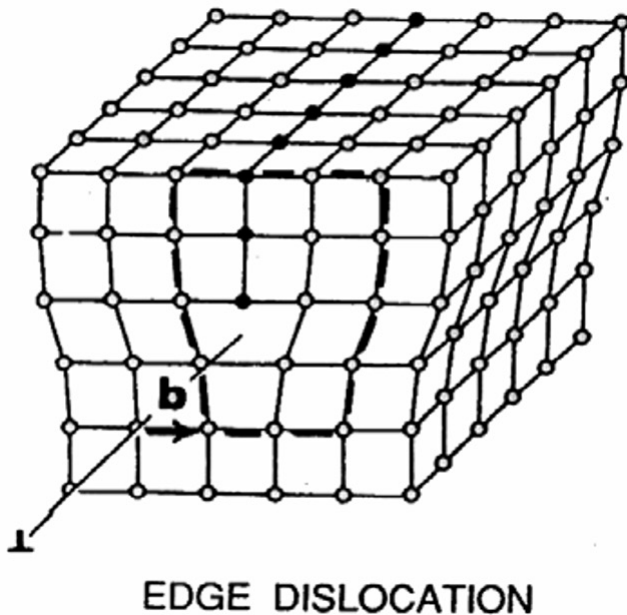
(d)

**Point defects dictate most dopant diffusion mechanisms,
and thus determine the impurity profile.**

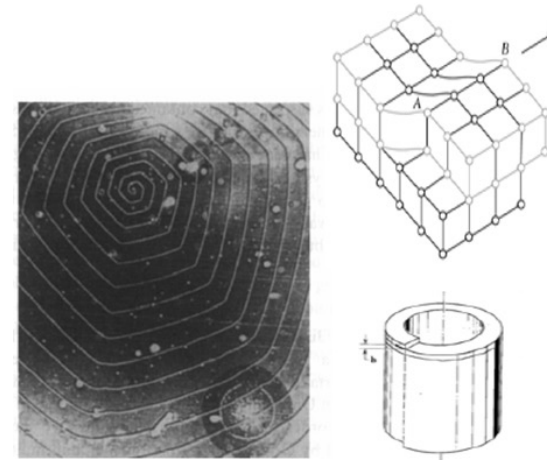
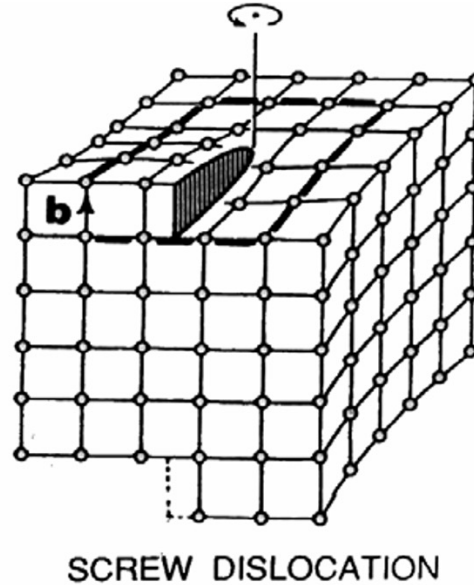
2.1.3 Linear defects

A missing line or additional line of atoms is called a dislocation

Edge dislocation

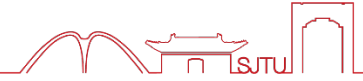


Screw dislocation

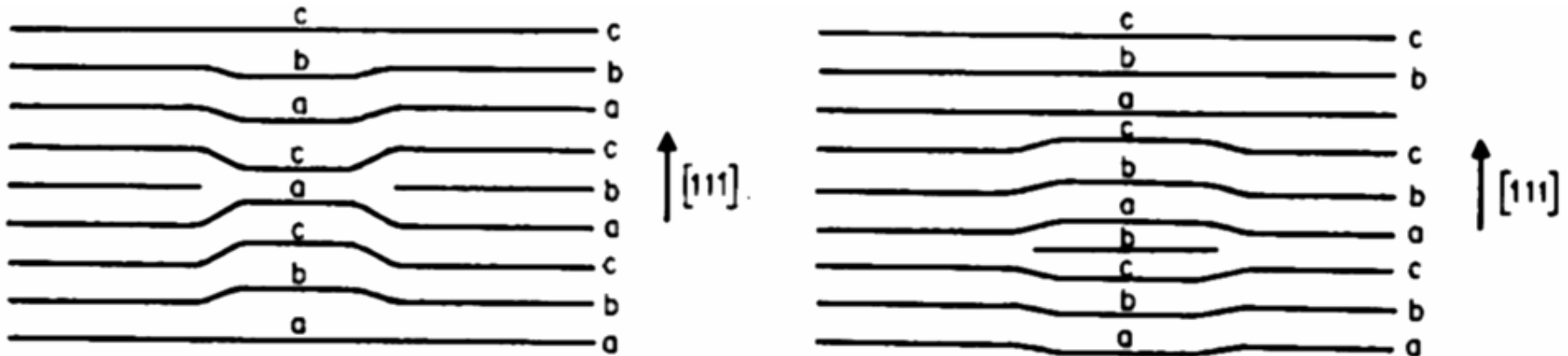


Screw dislocations give a helical structure to the planes, often show up in crystal growth

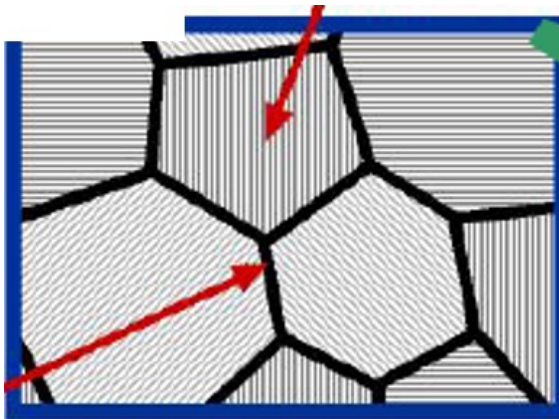
2.1.3 Planar defects



A stacking fault is a disruption in the stacking of layers in the crystal.
It is terminated by dislocation.

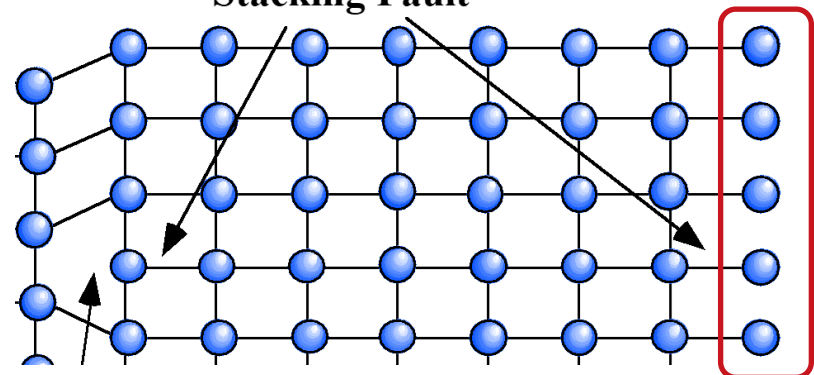


Grain boundary



Surface

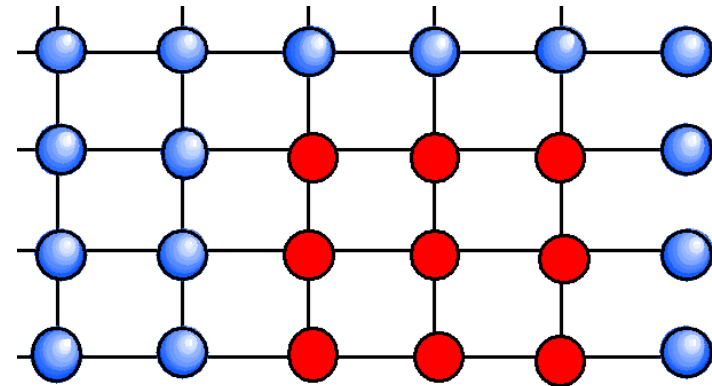
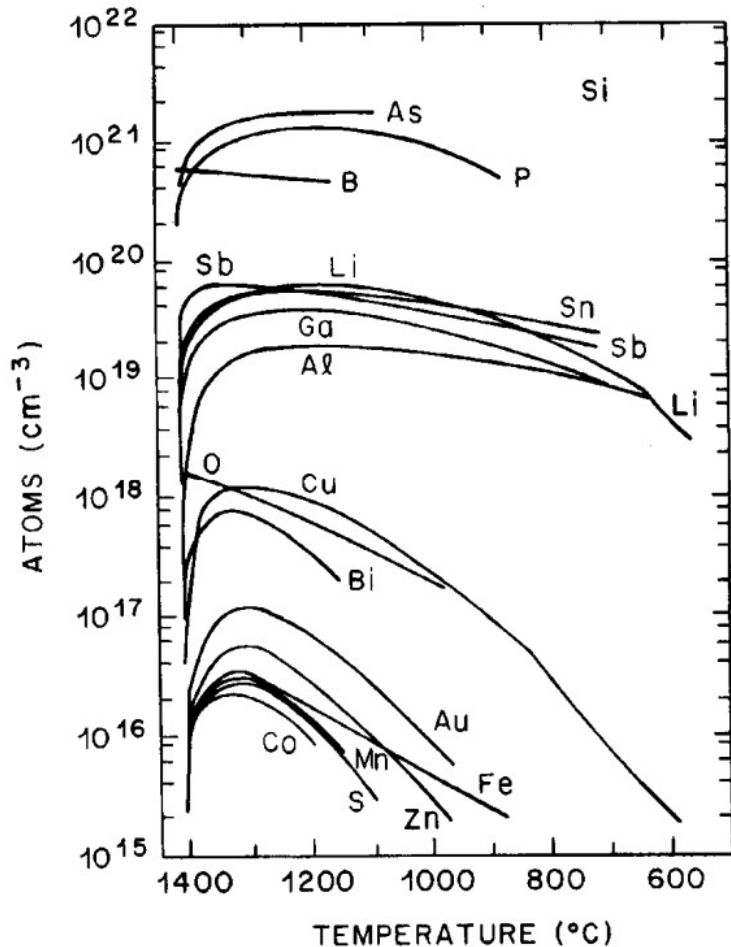
Stacking Fault



2.1.3 Bulk defects

Precipitates are three dimensional defects that have a different chemical makeup from the host lattice

Precipitation in Si



Most impurities have a solid solubility: If such an impurity is introduced at the maximum concentration allowed by its solubility at high temperature, and then cooled to a lower temperature, a supersaturated condition exists. Impurity atoms in excess of the solubility level will precipitate as a second phase.

✎ **2.1 Silicon crystal structure and defects**

✎ **2.2 Single crystal growth**

✎ **2.3 Wafer manufacturing and evaluation**

✎ **2.4 Wafer cleaning**

✎ **2.5 Summary**

2.2.1 Raw material for Si crystal .

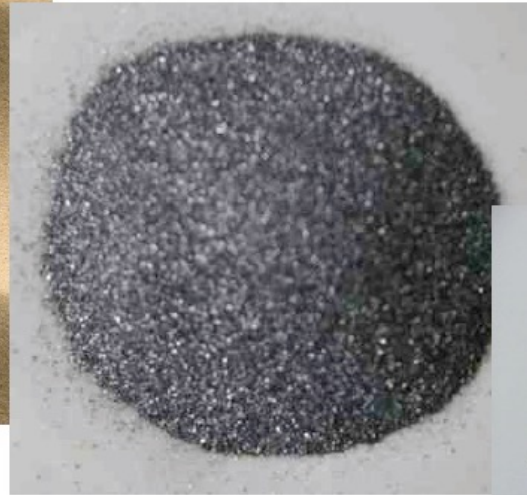


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Purification of MGS Silicon



Sand
天然砂



Metallurgical:冶金学
Metallurgical Grade Silicon
(MGS)



Electronics Grade Silicon
(EGS)

From Sand to MGS silicon



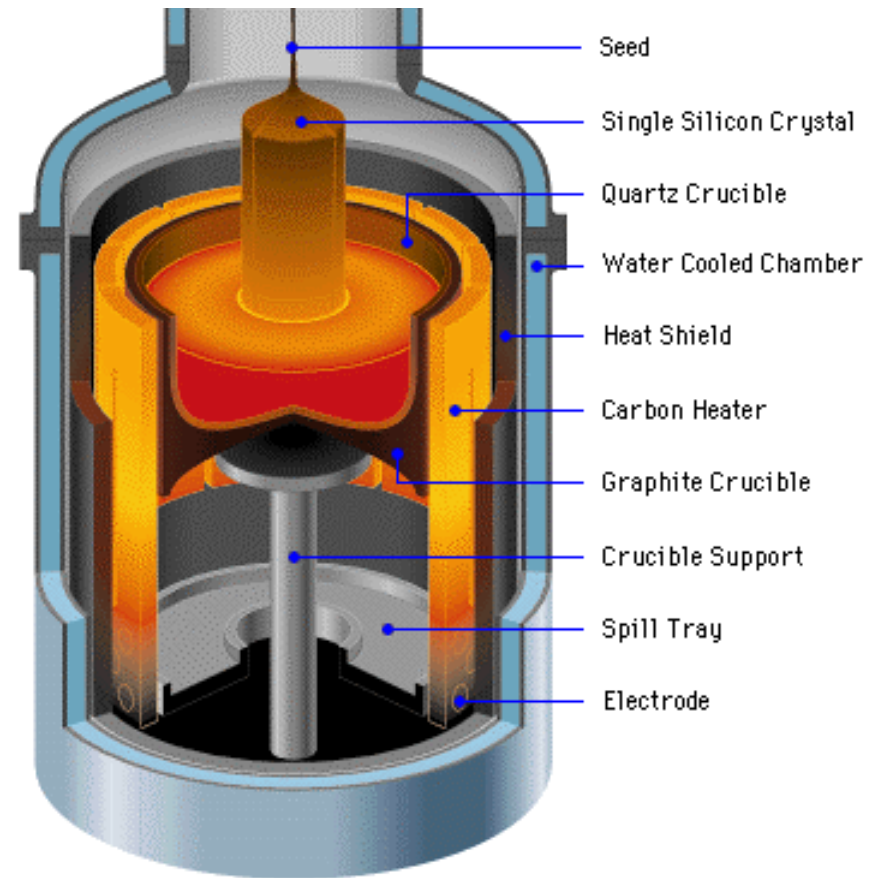
Raw material (EGS silicon) is very pure: < 1 ppb ($\sim 5 \times 10^{13} \text{ cm}^{-3}$) impurities except for O ($\sim 10^{18} \text{ cm}^{-3}$) and C ($\sim 10^{16} \text{ cm}^{-3}$)

2.2.2 Czochralski single crystal growth.



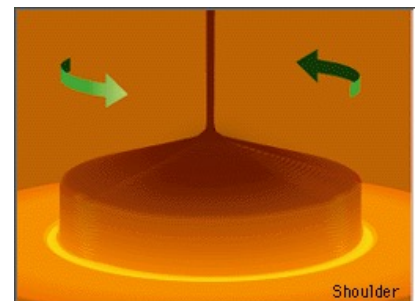
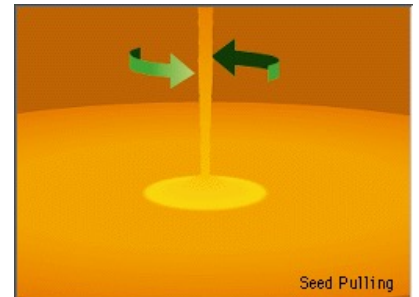
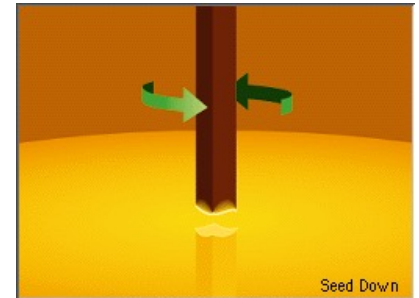
Virtually all Si wafers used for ICs today are produced by Czochralski (CZ) method

Jan Czochralski discovered the Czochralski method in 1916 when he accidentally dipped his pen into a crucible (坩埚) of molten tin rather than his inkwell (墨水盒). He immediately pulled his pen out to discover that a thin thread of solidified metal was hanging from the nib. And Czochralski verified that the crystallized metal was a single crystal.



2.2.2 Czochralski single crystal growth.

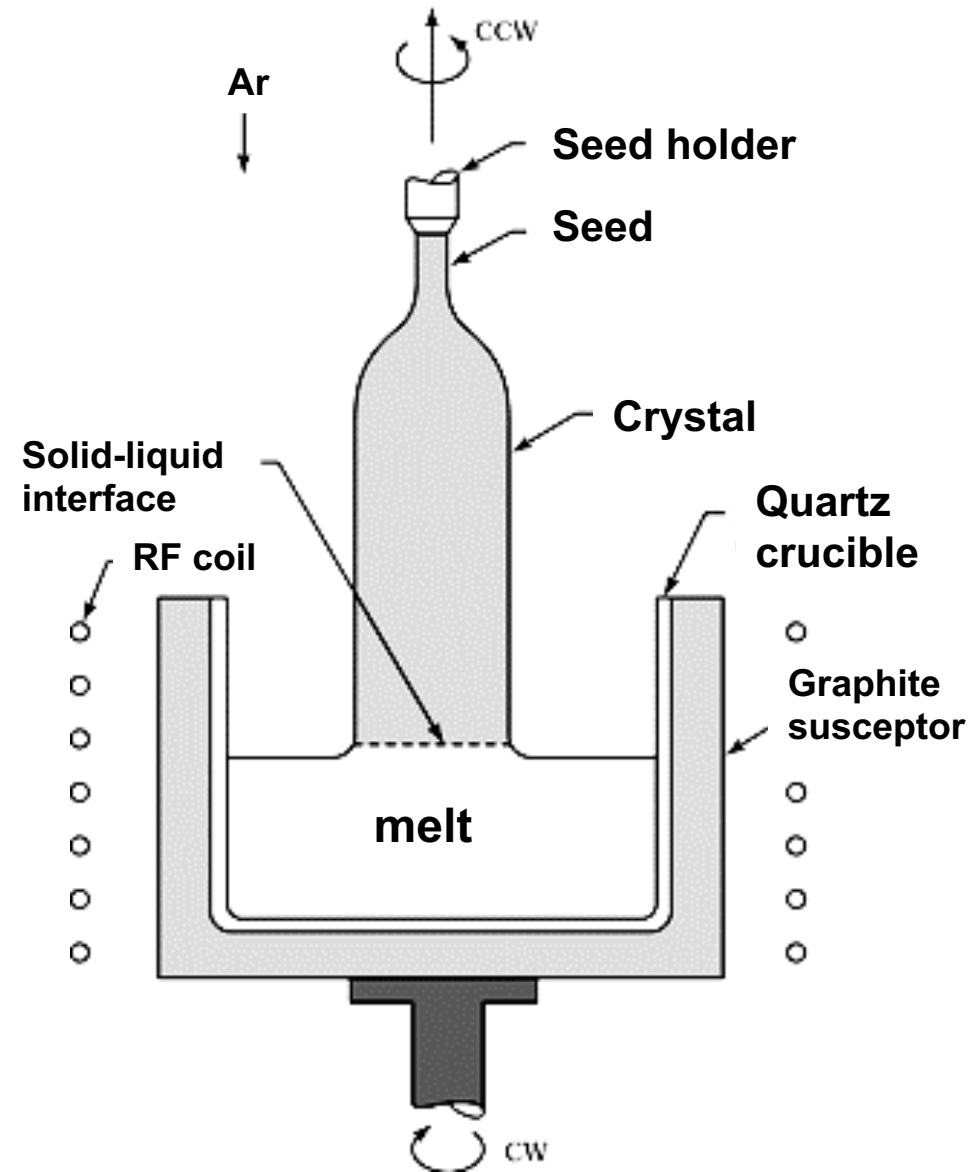
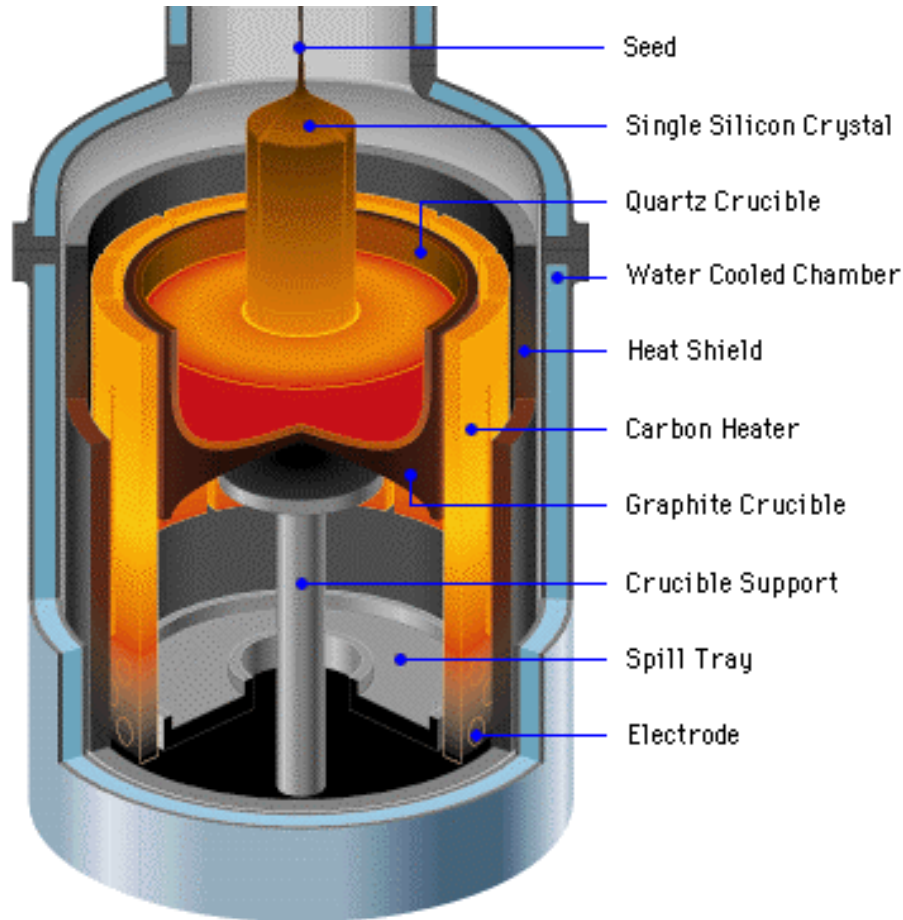
- EGS is broken into small pieces and placed in an SiO_2 crucible. In an Ar ambient, the crucible is heated to just above 1417°C .
- Dopant is added to the melt to intentionally dope the resulting crystal
- A single crystal seed is then lowered into the melt and withdrawn slowly. Melt flows up the seed and cools as crystal begins to grow.
- Seed rotated about its axis to produce a circular cross-section crystal. The rotation inhibits the natural tendency of the crystal to grow along certain orientations to produce a faceted crystal.
- Long ingots (錠) (boules) ~ 100 kg, with very good circular cross-section are produced.



Pull rate, melt temperature and rotation rate are all important control parameters.

2.2.2 Czochralski single crystal growth.

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2.2.2 Czochralski single crystal growth.

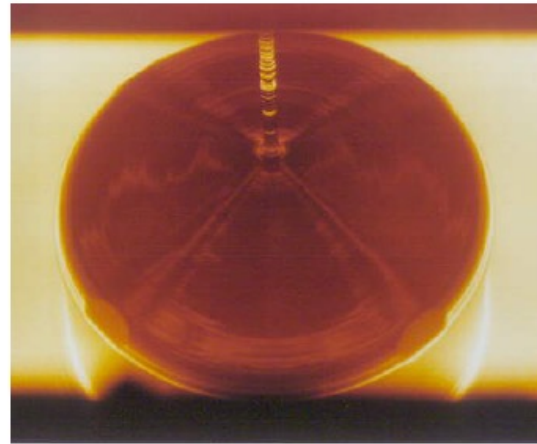


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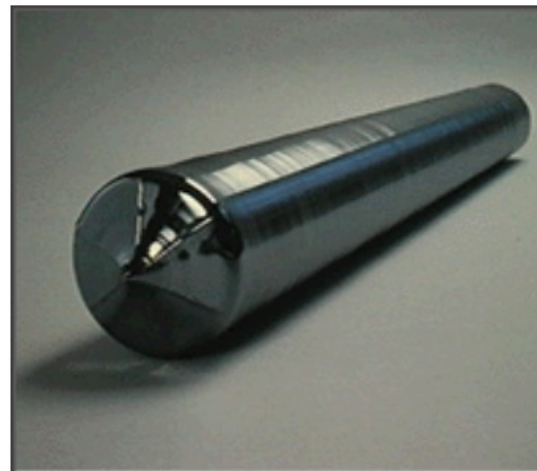
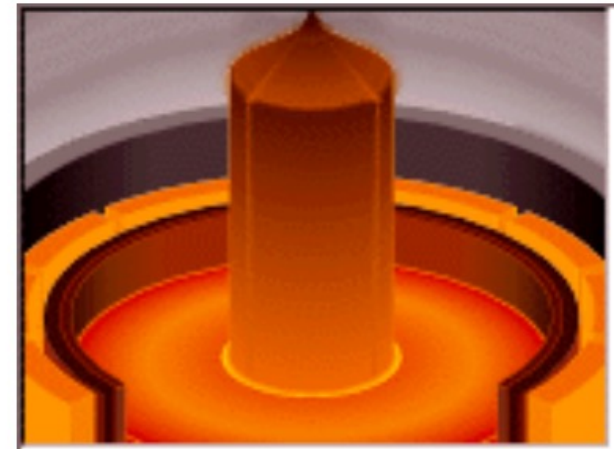
A commercial CZ puller



Early in the growth process



Later in the growth process

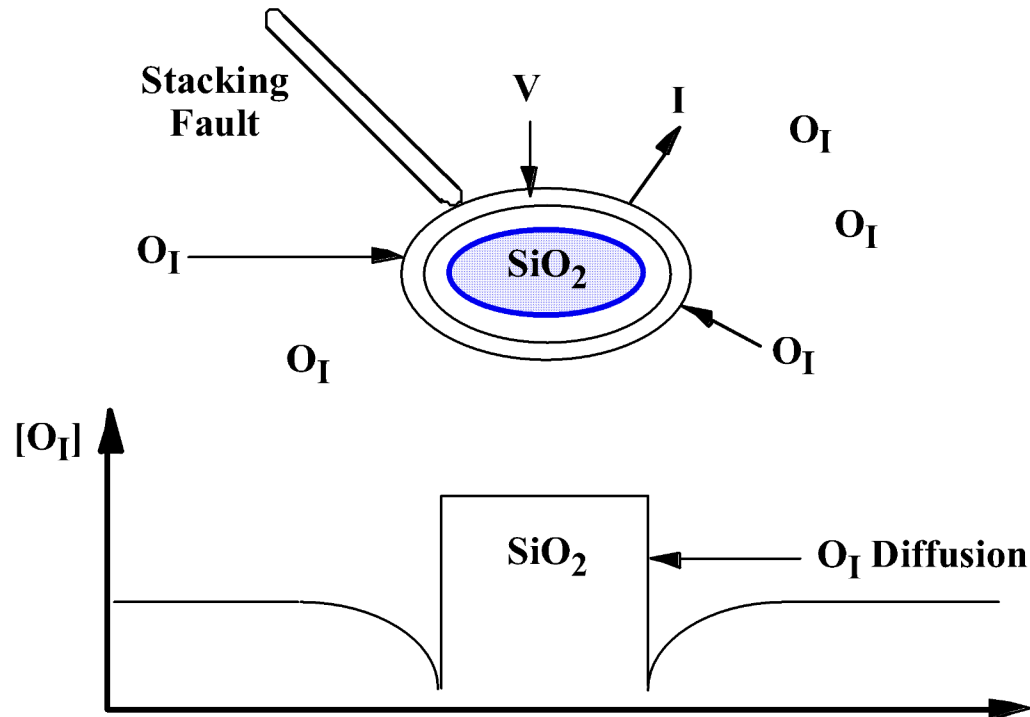


finished
boules

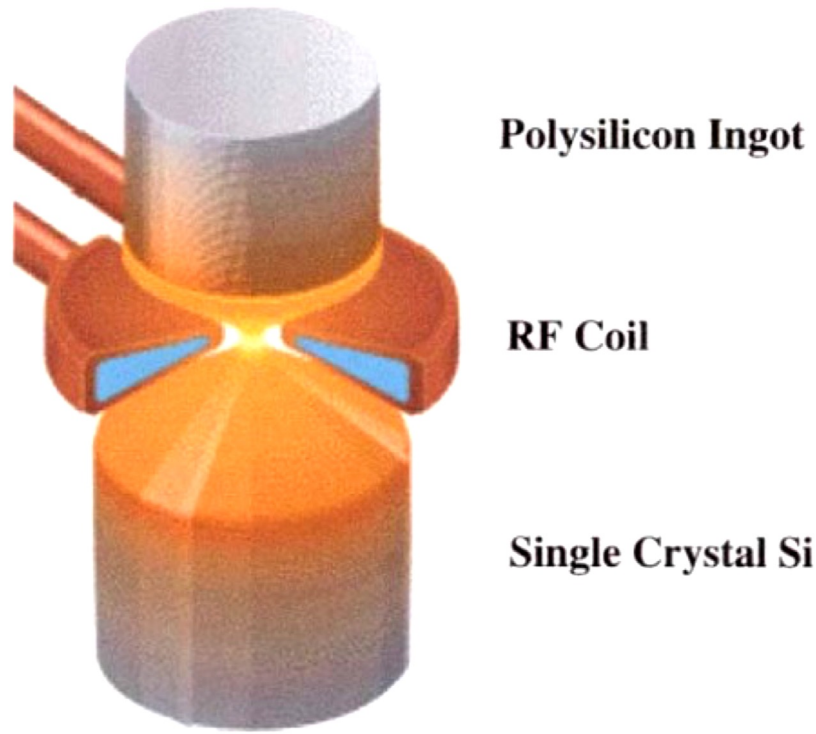
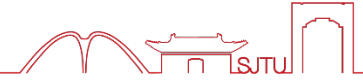


2.2.3 Oxygen and carbon in Si

- The CZ growth process inherently introduces O (from SiO_2 crucible) and C (from graphite susceptor/supporter).
- Typically, $C_O \approx 10^{18} \text{ cm}^{-3}$ and $C_C \approx 10^{16} \text{ cm}^{-3}$.
- The O in CZ silicon often forms small SiO_2 precipitates in the Si crystal under normal processing conditions.
- O and these precipitates can actually be very useful: provide mechanical strength, internal gettering (吸杂) .



2.2.4 Float Zone process



- An alternative growth process is the Float Zone method. It can be used for either refining (removing impurities) or single crystal growth.
- This process results in higher purity wafers which are typically used in power devices or concentrator solar cells.

✎ **2.1 Silicon crystal structure and defects**

✎ **2.2 Single crystal growth**

✎ **2.3 Wafer manufacturing**

✎ **2.4 Wafer cleaning**

✎ **2.5 Summary**

2.3.1 Main process of wafer manufacturing



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1. EGM
fabrication



2. Single crystal
growth



3. diameter
grinding



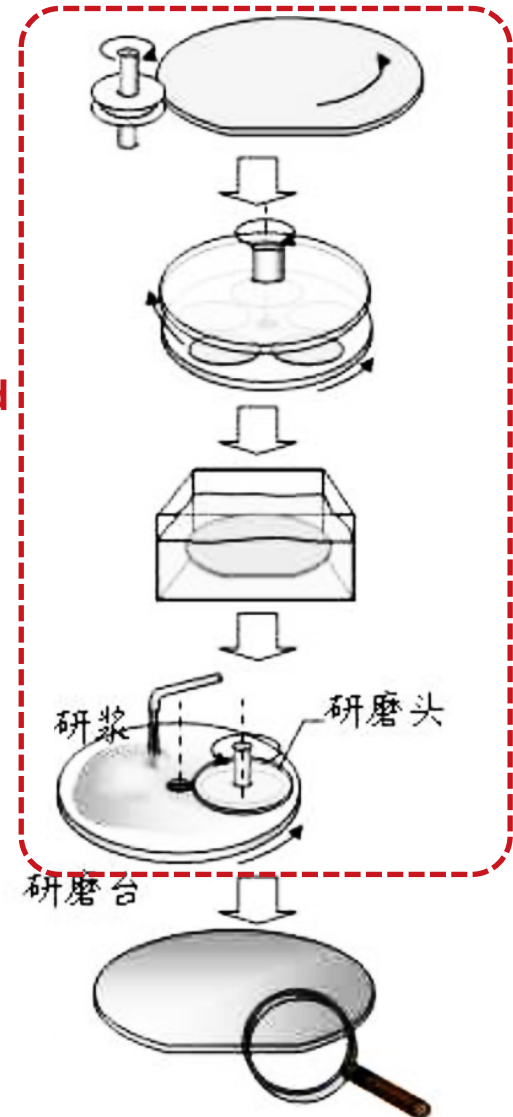
4. Main plane
formation



5. Slicing



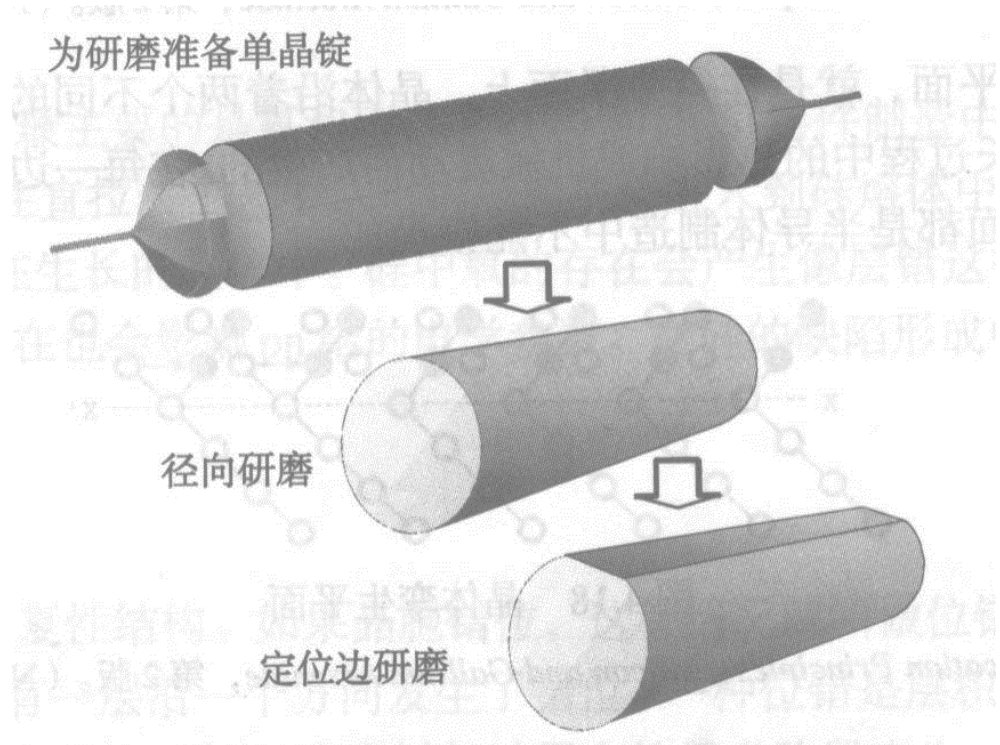
6. Wafer
polishing and
marking



7. Wafer
evaluation

Grinding

In silicon growth, there is deviation in diameter, so the silicon crystal needs to be barrelled (滚磨) into a target diameter



Orientation confirmation

Using X ray to confirm the target orientation and to grid a reference plane

2.3.2 Slicing of silicon wafer

- Cutting silicon wafer from crystal using diamond cutter

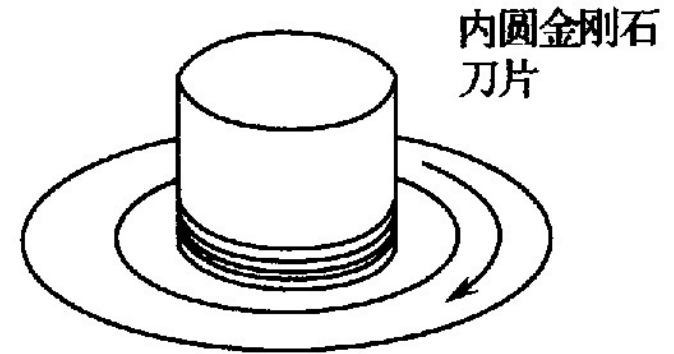
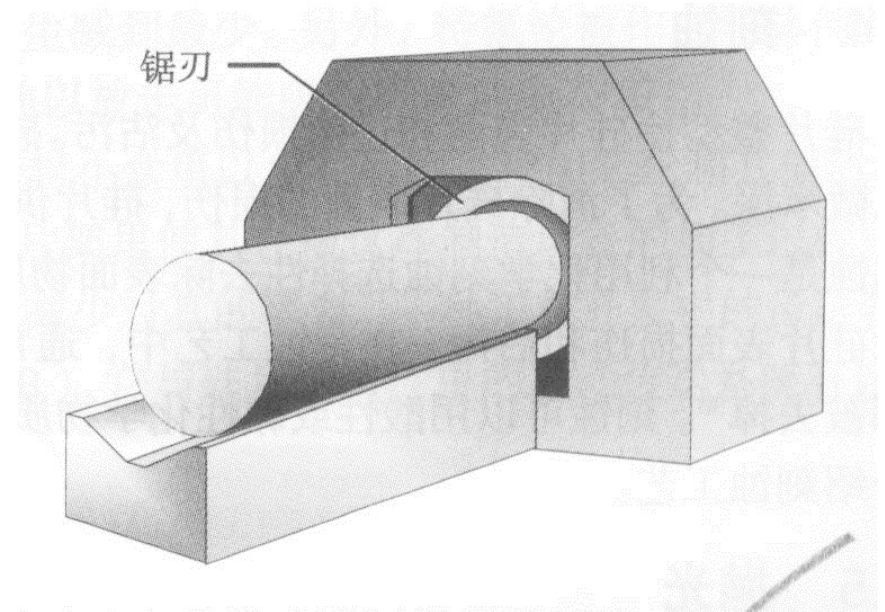
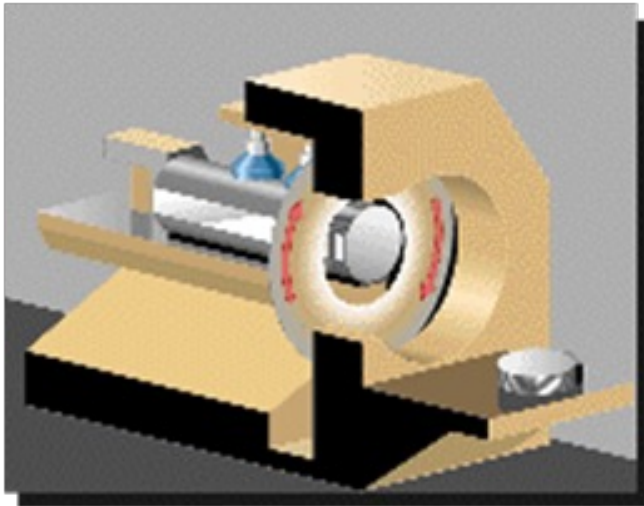
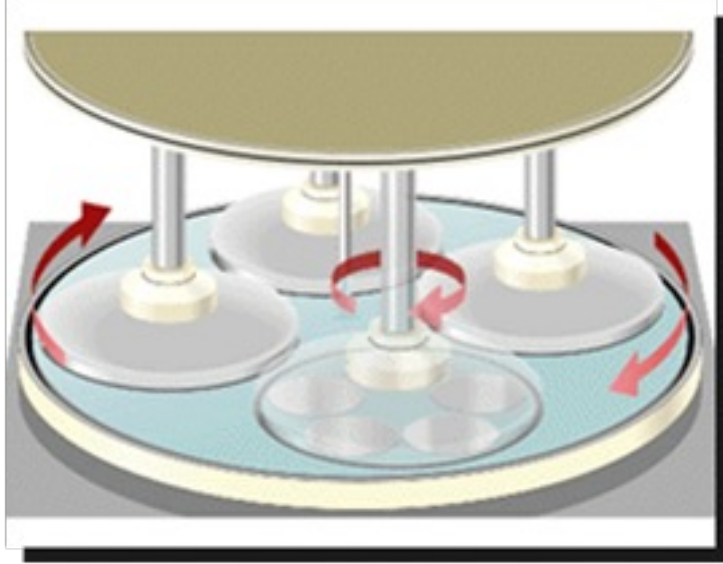


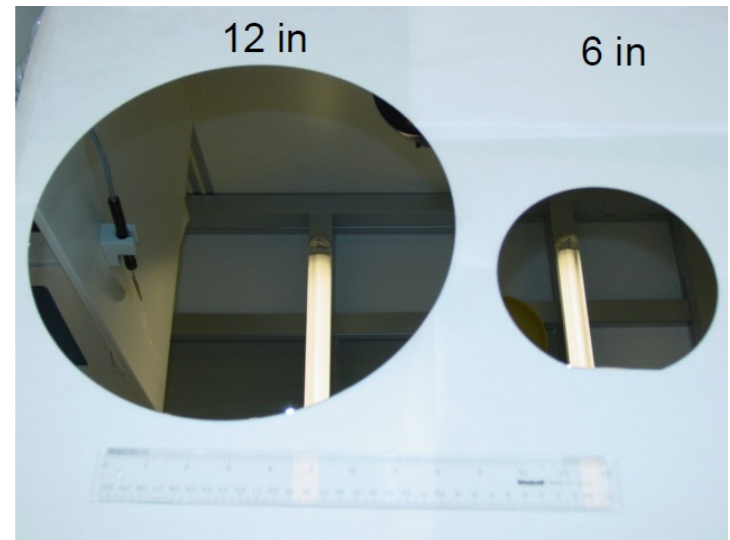
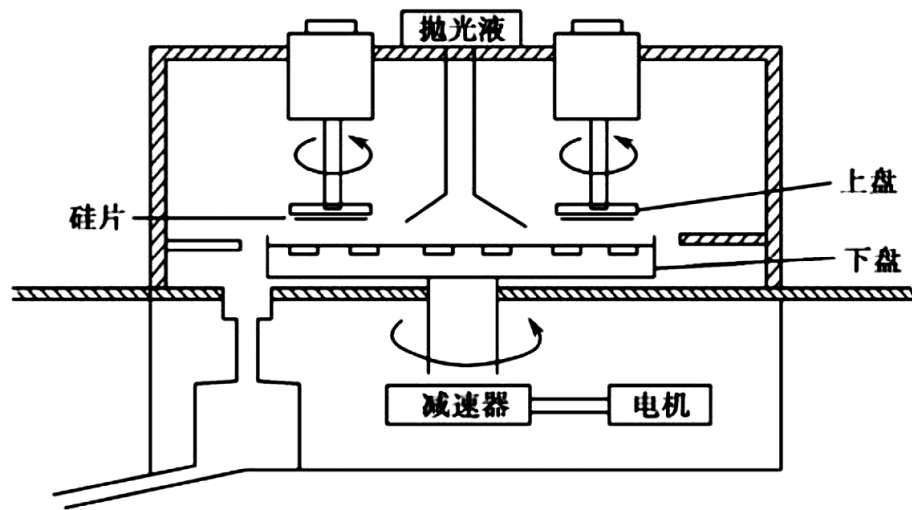
图 3.6 晶片切割



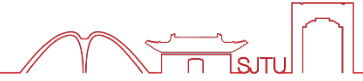
2.3.2 Polishing of Silicon wafer



- Smooth surface without damaging is required for silicon wafer for IC.
- Chemical-mechanical polishing(CMP)
- Curvature, roughness controlling
- Marking information using laser

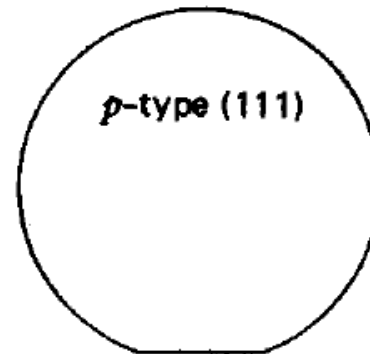


2.3.3 Orientation of Silicon wafer

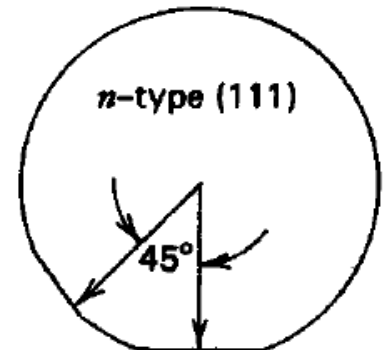


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Flat patterns reveal
doping type and in-plane
crystal orientation

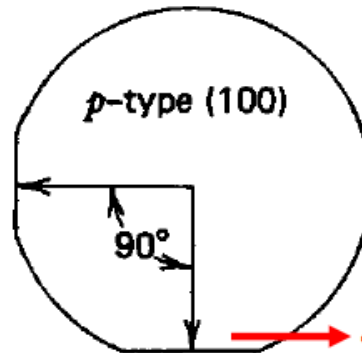


Orientation flat on
(1 $\bar{1}$ 0) plane



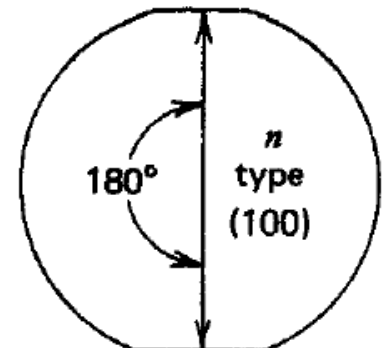
Orientation flat on
(1 $\bar{1}$ 0) plane

Most
common



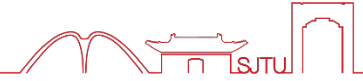
Orientation flat on
(110) plane

$\langle 110 \rangle$



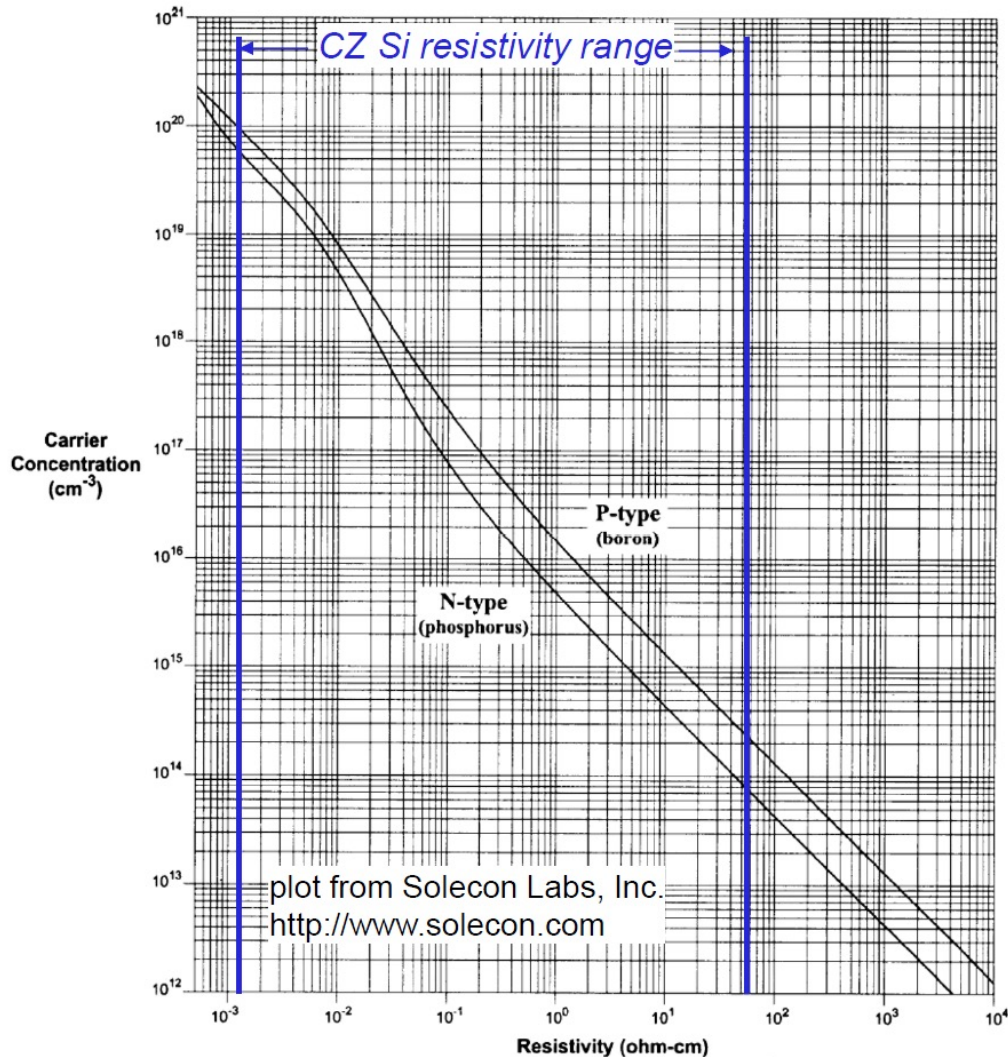
Orientation flat on
(110) plane

2.3.3 Resistivity of Silicon wafer



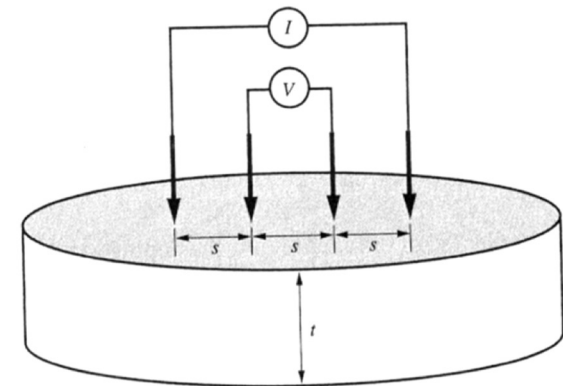
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Resistivity vs. Doping



$$\rho = \frac{1}{q\mu_n n + q\mu_p p} (\Omega\text{-cm})$$

Four-point probe



✎ **2.1 Silicon crystal structure and defects**

✎ **2.2 Single crystal growth**

✎ **2.3 Wafer manufacturing and evaluation**

✎ **2.4 Wafer cleaning**

✎ **2.5 Summary**

2.4.1 Why wafer cleaning is needed?



Example 1: MOSFET threshold voltage is given by:

$$V_T = \phi_{ms} - \frac{Q_{ox}}{C_{ox}} + 2\psi_b + \frac{\sqrt{2\epsilon_s q N_A (2\psi_b)}}{C_{ox}}$$

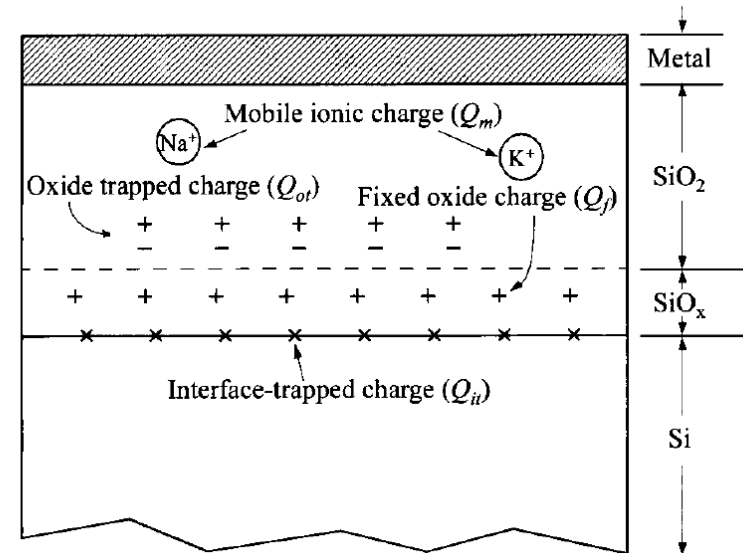
Where Q_M is the concentration of mobile ions (e.g. Na^+ and K^+ in particular) in the oxide, expressed as charges per cm^2

$$q = 1.6 \times 10^{19} \text{ cm}^{-3}$$

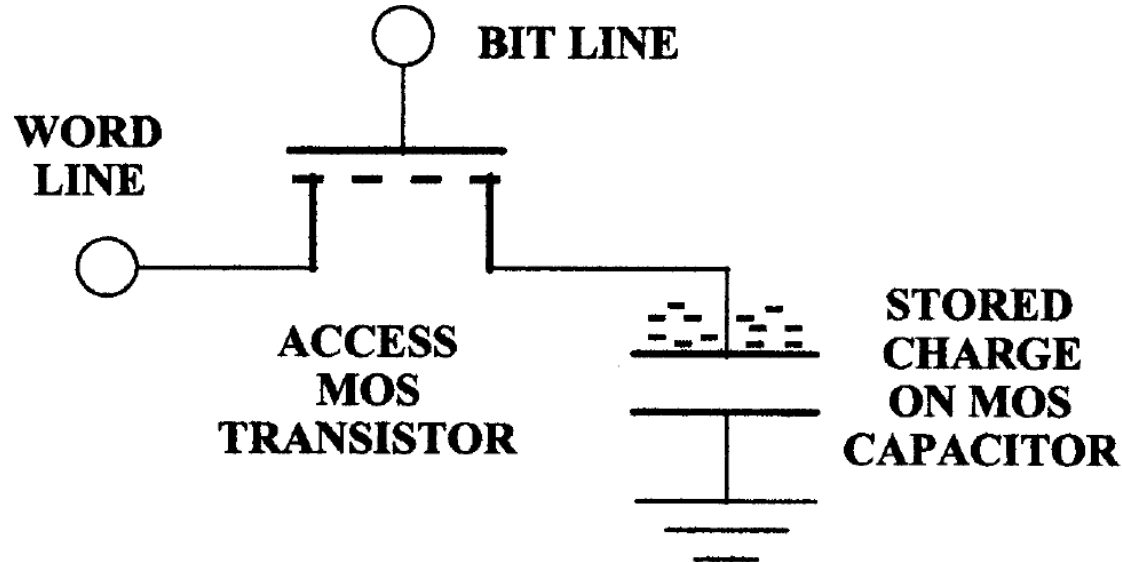
$$C_o = \frac{\epsilon_{ox}}{t_{ox}}$$

$$\epsilon_{ox} = (3.9)^* (8.85 \times 10^{-14}) = 3.45 \times 10^{-13} \text{ F/cm}$$

If $t_{ox} = 10 \text{ nm}$, then a $0.1 \text{ Volt } V_{th}$ shift can be caused by $Q_M = 2.2 \times 10^{11} \text{ cm}^{-2}$ ($< 0.1\%$ monolayer in the oxide)



Example 2: MOS DRAM



- Charge leaks out of the MOS capacitor, and must be refreshed
- Refresh time of several msec requires a generation lifetime $\tau_G > 25 \mu\text{sec}$

$$\tau_G = \frac{1}{\sigma v_{th} N_t}$$

$$v_{th} \sim 10^7 \text{ cm/s}$$
$$\sigma \sim 10^{-15} \text{ cm}^2$$

$$\tau_G \sim 100 \text{ } \mu\text{sec} \rightarrow N_t \sim 10^{12} \text{ cm}^{-3}$$

2.4.1 Why wafer cleaning is needed?

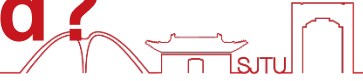


Projected Defect and Contamination Requirements

Year of 1st DRAM Shipment	1997	1999	2003	2006	2009	2012
Minimum Feature Size	250nm	180nm	130nm	100nm	70nm	50nm
Wafer Diameter (mm)	200	300	300	300	450	450
DRAM Bits/Chip	256M	1G	4G	16G	64G	256G
DRAM Chip Size (mm ²)	280	400	560	790	1120	1580
Microprocessor Transistors/chip	11M	21M	76M	200M	520M	1.40B
Critical Defect Size	125nm	90nm	65nm	50nm	35nm	25nm
Starting Wafer Total LLS (cm⁻²)	0.60	0.29	0.14	0.06	0.03	0.015
DRAM GOI Defect Density (cm⁻²)	0.06	0.03	0.014	0.006	0.003	0.001
Logic GOI Defect Density (cm⁻²)	0.15	0.15	0.08	0.05	0.04	0.03
Starting Wafer Total Bulk Fe (cm⁻³)	3x10¹⁰	1x10¹⁰	Under 1x10¹⁰	Under 1x10¹⁰	Under 1x10¹⁰	Under 1x10¹⁰
Critical Metals on Wafer Surface After Cleaning (cm⁻²)	5x10⁹	4x10⁹	2x10⁹	1x10⁹	< 10⁹	< 10⁹
Starting Material Recombination Lifetime (μsec)	≥ 300	≥ 325	≥ 325	≥ 325	≥ 450	≥ 450

From ITRS website

2.4.1 Why wafer cleaning is needed?



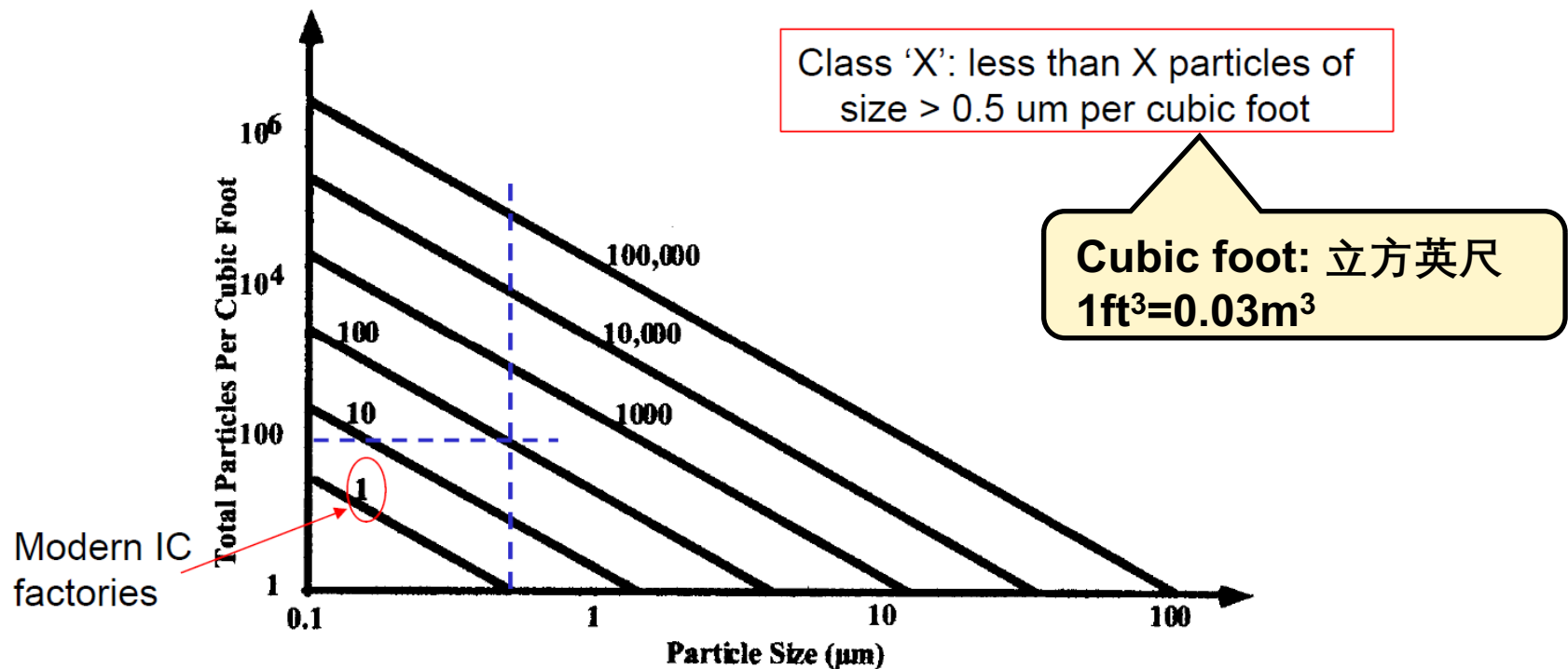
- Such traps are usually associated with deep-level impurities like Au, Cu, Fe and other elements;
- The metallic impurity concentration must be very low;
- In fact, the requirements are even more severe, because elements like Au, Cu and Fe tend to preferentially accumulate at the wafer surfaces and in junction regions of devices
- Impossible to keep the concentration of all impurities below these levels

==> use facilities and processes

Cleaning room + Cleaning process

2.4.2 Cleaning factories

Air quality is measured by the 'class' of the facility



Particle size distributions for various classes of clean rooms.

Vertical axis is the total number of particles larger than a given size.

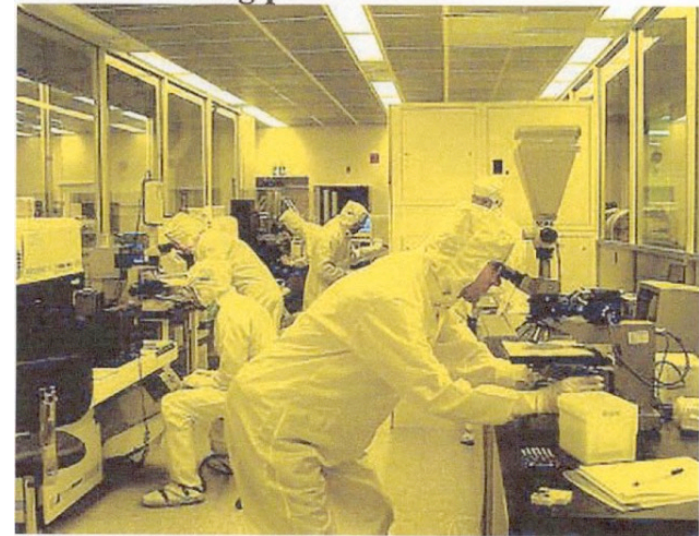
2.4.2 Cleaning factories

Factory environment is cleaned by:

- HEPA filters and re-circulation of the air

HEPA(High efficiency particulate air)

- “Bunny suits” or “space-suits” for workers
- Filtration of water, chemicals and gases



University-style cleanroom



200 mm industrial fab



AEMD in SJTU

AEMD平台分别座落于交大闵行校区微电子大楼一层（西区）以及综合实验楼一层（东区）（2014年8月东西区合并），实验室共有近1510m²的100级、1000级净化室，其中100级210 m²，1000级约1300m²。

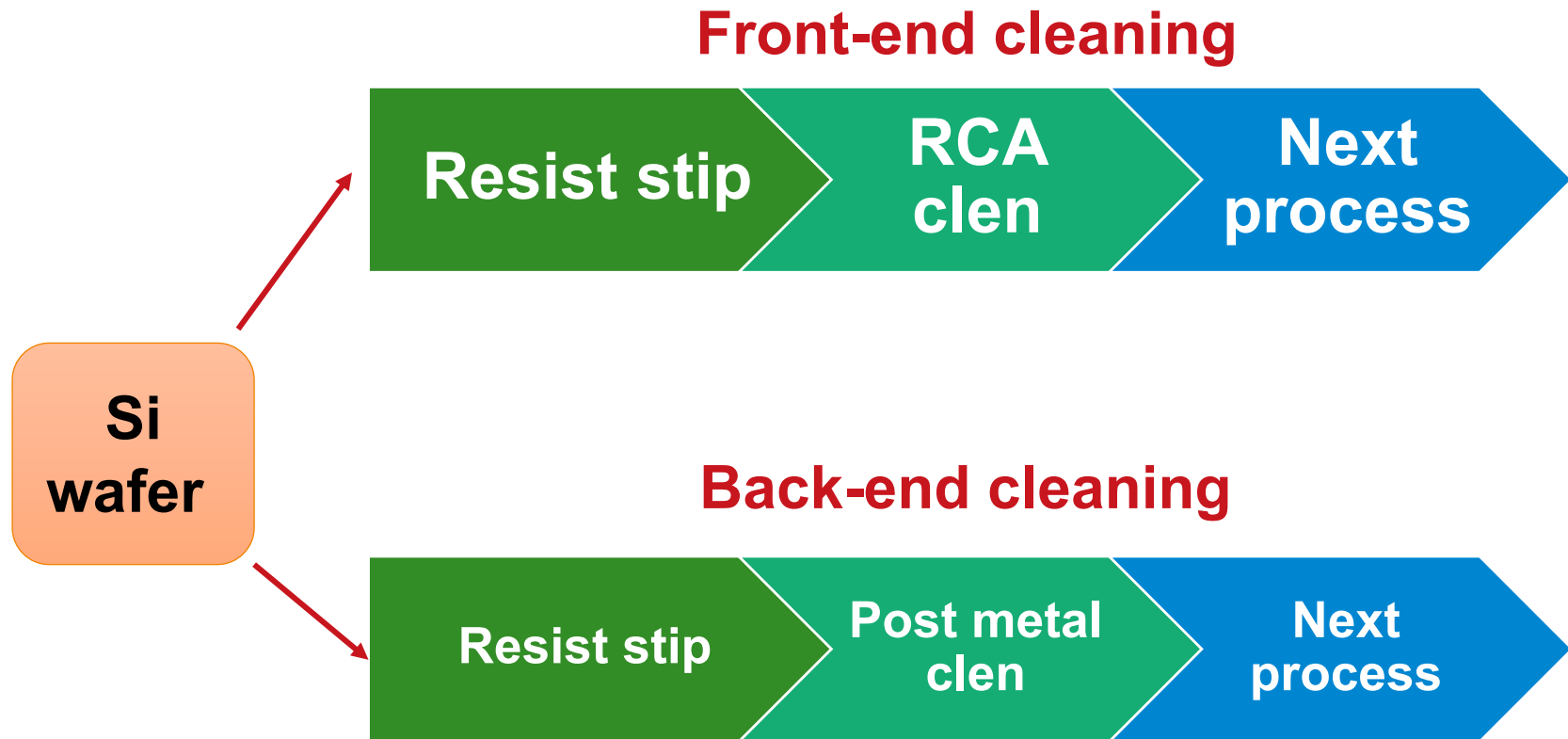


上海交通大学
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先进电子材料与器件校级平台
Center for Advanced Electronic Materials and Devices

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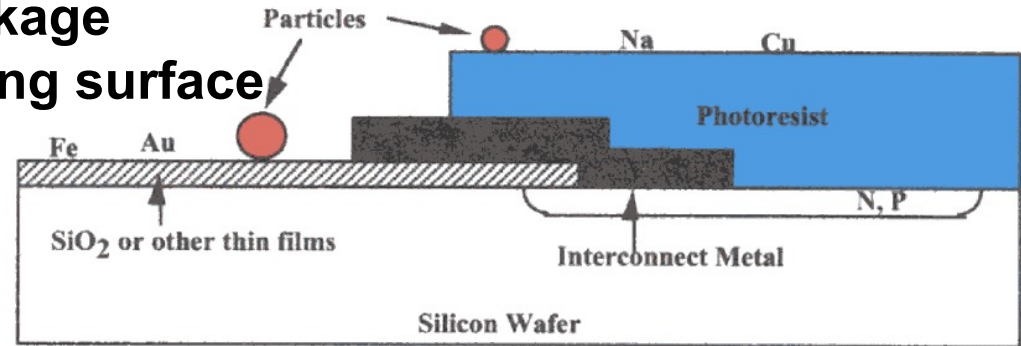


- Front-end cleaning is performed prior to each temperature step
- Back-end cleaning uses less caustic solutions since at that point in the flow, metal wiring is on the wafer

2.4.3 Contaminations

(1) Organic films or trace organics

- can cause increased gate leakage
- prevents cleaning of underlying surface



(2) Particles

- same reason as above as well as killer defects
- may carry metals from the handling equipment
- can cause reliability problems in gate oxides

(3) alkali metals (Ca, Na)

- move easily in oxides in response to an applied field
- cause oxide reliability problems and V_t shifts

(4) transition metals (Fe, Au, Cu)

- diffuse readily into the wafer during temperature steps
- can cause surface micro-roughness
- reduce carrier lifetime and mobility

➤ Dry cleaning

Oxygen plasma

Decompose resistor to $\text{CO}_2 + \text{H}_2\text{O}$

➤ Wet cleaning

$\text{H}_2\text{SO}_4(98\%): \text{H}_2\text{O}_2(30\%) = 2:1-4:1$

Decompose organic to $\text{CO}_2 + \text{H}_2\text{O}$

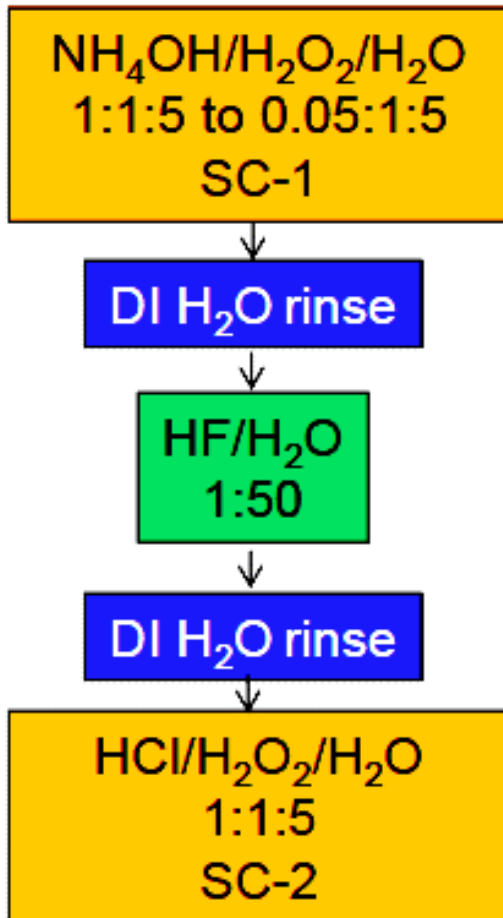
Organic cleaning is prior to the main or 'pre-diffusion' RCA wet chemical clean of the wafer. Remaining trace organics are removed in the RCA clean, along with metals.

2.4.3 RCA clean



RCA cleaning is “Standard process” used to remove organics, heavy metals and alkali ions.

Werner Kern developed the basic procedure in 1965 while working for RCA, the Radio Corporation of America.



➤ **SC-1 (APM: Ammonia Peroxide Mixture)**

-70-80° C/10min

-Remove organics, and particles

-Etches Si surface very slowly

➤ **Remove oxide**

-Room T/1min

➤ **SC-2 (HPM: Hydrochloric Peroxide Mixture)**

-70-80° C/10min

-Remove alkali ions such as Al³⁺, Fe³⁺, Mg²⁺

-Remove heavy metals like Au

Front-end cleaning

	化学溶剂	清洗温度	清除的污染物
1	$\text{H}_2\text{SO}_4 + \text{H}_2\text{O}_2 (4:1)$	120°C, 10min	有机污染物
2	D.I. H_2O	室温	
3	$\text{NH}_4\text{OH} + \text{H}_2\text{O}_2 + \text{H}_2\text{O}$ (1:1:5) (SC-1)	80°C, 10min	微尘
4	D.I. H_2O	室温	
5	$\text{HCl} + \text{H}_2\text{O}_2 + \text{H}_2\text{O}$ (1:1:6) (SC-2)	80°C, 10min	金属离子
6	D.I. H_2O	室温	
7	$\text{HF} + \text{H}_2\text{O} (1:50)$	室温	氧化层
8	D.I. H_2O	室温	
9			干燥

2.4.3 Cleaning facilities



- ✎ 2.1 Silicon crystal structure and defects
- ✎ 2.2 Single crystal growth
- ✎ 2.3 Wafer manufacturing and evaluation
- ✎ 2.4 Wafer cleaning
- ✎ **2.5 Summary and homework**

- **Silicon crystal structure and basic properties**
- **Four kinds of defects in Si crystal structure**

Form, effect

- **CZ growth of silicon crystal**

Raw materials, physics, process, facilities

- **Silicon wafer fabrication and evaluation**

Process, tools, reasons, requirements

- **Contamination control of Silicon wafer**

Why, clean rooms, RCA cleaning:

physics and process

**用一张表总结晶圆制造（包括原材料和单晶生长）
和清洗的所有流程和目的**

**Summary all the processes and their
purpose of wafer manufacturing and
cleaning using a diagram or table**

单击此处编辑母版标题样式



谢谢

